Hugh O'Shaughnessy

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Education

University of Wisconsin–Madison Madison, WI

B.S. Computer Science, May 2026

Selected coursework: Foundation Models, Database Systems, Big Data Systems, Data Science, Computer Architecture, High-Performance Computing, Operating Systems, Compilers, Computer Networks, Algorithms, Data Structures.

Experience

ShiftIQ 2026–Sep 2026

Research Scientist Intern *Remote*

- Founding research team building governed AI infrastructure around the Causal Execution Engine (CEE).
- Working at the boundary between mechanistic interpretability and engine design: probing residual-stream representations in open-weight models to identify which operations the engine should preserve and govern.
- Running PyTorch hook experiments (activation patching, linear probes, causal interventions) to map which operations a model can reliably execute under steering, feeding results into the engine's design decisions.

Summer AI Lab (SAIL) Fellow, UW–Madison CDIS Jun–Aug 2025

N+1 Institute, OpenAI-Funded, Inaugural Cohort *Madison, WI*

- Selected for the inaugural Summer AI Lab cohort; received a $1,500 research scholarship and OpenAI API credits.
- Co-built a personalized agentic news aggregator on a 3-person research team: web-scraping ingestion, RAG retrieval, and source-conflict adjudication.
- Implemented semantic-similarity and bias-detection modules feeding a real-time credibility-scoring pipeline.

Projects

16-bit 5-Stage Pipelined Processor *Verilog, ModelSim*

- Co-built a team course project: a 16-bit, five-stage pipelined processor with IF/ID/EX/MEM/WB stages.
- Co-developed operand forwarding plus hazard and stall control, with branch and jump resolution in the decode stage.
- Co-developed a separate two-way cache controller and corresponding test harness; the final submitted processor used a stalled-memory interface.

MiniSpark: Parallel Data-Processing Framework *C, Unix*

- Multi-threaded analytics engine in C modelled on Apache Spark; runs parallel pipelines through a custom thread pool and DAG scheduler.
- Mutex-guarded shared state and a low-overhead monitor thread tracking per-stage performance and queue depth.
- Implemented core operators (map, filter, join) with stage-boundary materialization and dependency-aware scheduling.

Student Enrollment DBMS *SQL, C++*

- Normalized SQL schema (students, professors, courses, enrollments) with referential integrity across join tables; ER model validated against query-performance tests.
- C++ application layer exposing CRUD operations and parameterized query helpers over the database.

Technical Skills

Languages: Python, TypeScript, SQL, C, C++, Verilog, Java

AI / LLM: PyTorch, Hugging Face Transformers, LangGraph, OpenAI / Anthropic APIs, pgvector, RAG, agentic workflows, model evaluation, post-training quantization

Data / Backend: PostgreSQL, MongoDB, Snowflake, Elasticsearch, Apache Spark, Airtable, Node.js, REST APIs, ETL / data pipelines

Web / Infrastructure / EDA: React, Next.js, Linux, Docker, AWS, CI/CD (GitHub Actions), Git, ModelSim, Quartus, Xilinx/AMD Versal